

01. පහත සඳහන් සීමා කොයන්න.

i.  $\lim_{x \rightarrow a} \left( \frac{x^{100} - a^{100}}{x - a} \right)$     ii.  $\lim_{x \rightarrow -2} \left( \frac{x^3 + 8}{x + 2} \right)$     iii.  $\lim_{x \rightarrow 2} \left( \frac{\frac{1}{x^2} - \frac{1}{4}}{x - 2} \right)$     iv.  $\lim_{x \rightarrow 1} \left( \frac{x^{\frac{3}{4}} - 1}{1 - x^{\frac{2}{3}}} \right)$     v.  $\lim_{x \rightarrow 1} \left( \frac{x^2 - x^{-2}}{x^5 - x^{-5}} \right)$

02. පහත සඳහන් සීමා කොයන්න.

i.  $\lim_{x \rightarrow 0} \frac{\tan x}{x}$     ii.  $\lim_{x \rightarrow 0} \frac{2x - \sin x}{2x + \sin x}$     iii.  $\lim_{x \rightarrow 0} \frac{\sin 5x \cos 3x}{x}$     iv.  $\lim_{x \rightarrow 0} \frac{x + \sin 3x}{x - \tan 3x}$   
v.  $\lim_{x \rightarrow 0} \frac{\tan 3x \sec 2x}{4x}$     vi.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cot x - 1}{\cos 2x}$     vii.  $\lim_{x \rightarrow 0} \frac{1 - \cos^2 2x}{x^2}$     viii.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{\cos x - \sin x}$   
ix.  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$     x.  $\lim_{x \rightarrow 0} \frac{1 - \cos 3x + 3x}{\sin 3x}$     xi.  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{4x^2 - \pi^2}$     xii.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sin 2x}{(\cos x - \sin x) \left( x - \frac{\pi}{4} \right)}$

03. i.  $\lim_{x \rightarrow \infty} \frac{ax^2 + bx + c}{lx^2 + mx + n}$  කොයන්න.

04. i.  $\lim_{x \rightarrow \infty} \frac{x^4 + 3x + 3}{5x^2 + 1}$     ii.  $\lim_{x \rightarrow \infty} \frac{2x + 3}{x^2 + 5x - 1}$     iii.  $\lim_{x \rightarrow \infty} \frac{2x^3}{x^2}$     iv.  $\lim_{x \rightarrow \infty} \frac{x^3 + 5x - 1}{2x^2 + 7}$  යන සීමා කොයන්න.

05. පහත සඳහන් සීමා කොයන්න.

i.  $\lim_{x \rightarrow 3} \left( \frac{x - 3}{\sqrt{x + 1} - 2} \right)$     ii.  $\lim_{x \rightarrow 4} \left( \frac{\frac{1}{\sqrt{x}} - \frac{1}{2}}{x - 4} \right)$     iii.  $\lim_{h \rightarrow 0} \left( \frac{\sqrt{x + h} - \sqrt{x}}{h} \right)$     iv.  $\lim_{x \rightarrow 0} \left( \frac{x}{\sqrt{1 + x} - 1} \right)$   
v.  $\lim_{x \rightarrow 1} \left( \frac{x^2 - 1}{\sqrt{3x + 1} - \sqrt{5x - 1}} \right)$

06. පහත සඳහන් සීමා පවති දැයි කොයා පවතින සීමාවන් කොයන්න.

i.  $\lim_{x \rightarrow 7} \frac{\sqrt{x + 2}}{x + 2}$     ii.  $\lim_{x \rightarrow 1} \frac{\sqrt{x + 1}}{\sqrt{x} - 1}$     iii.  $\lim_{x \rightarrow 0} \frac{x^2 + i}{\sqrt{x}}$     iv.  $\lim_{x \rightarrow 3} \frac{\sqrt{3x - 3}}{x^2 - 9}$     v.  $\lim_{x \rightarrow -1} \frac{\sqrt{3 - x} - 2}{x + 1}$   
vi.  $\lim_{x \rightarrow 0} x^{\frac{3}{2}}$     vii.  $\lim_{x \rightarrow -1} x^{\frac{2}{3}}$     viii.  $\lim_{x \rightarrow 0} \frac{x^{\frac{3}{5}} - x}{2x}$     ix.  $\lim_{x \rightarrow 1} \frac{x^{\frac{2}{3}} + 1}{x - 1}$     x.  $\lim_{x \rightarrow 1} \frac{x^4 - x}{\sqrt{7x + 9} - 4}$   
xi.  $\lim_{x \rightarrow 2} \frac{\sqrt{2x + 1} - \sqrt{5}}{x - 2}$

07. පහත සඳහන් සීමා කෙටීමට  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$ ,  $n \in \mathbb{Q}$  යන ප්‍රමේයය භාවිත කරන්න.

i.  $\lim_{x \rightarrow 2} \frac{x^2 - 8}{x - 2}$     ii.  $\lim_{x \rightarrow 1} \frac{x^{-5} - 1}{x - 1}$     iii.  $\lim_{x \rightarrow 81} \frac{\sqrt{x} - 9}{x - 81}$     iv.  $\lim_{x \rightarrow 64} \frac{\sqrt[3]{x} - 4}{\sqrt{x} - 8}$

08. පහත සඳහන් සීමාවන් සොයන්න.

i.  $\lim_{x \rightarrow \infty} \frac{ax^2 + bx + c}{cx^2 + bx + a}$

ii.  $\lim_{x \rightarrow \infty} \frac{2x}{x^2 + 1}$

iii.  $\lim_{x \rightarrow \infty} \frac{2x^2}{x^2 + 1}$

iv.  $\lim_{x \rightarrow \infty} \frac{2x^3}{x^2 + 1}$

v.  $\lim_{x \rightarrow \infty} \frac{5 - x^2}{x^2}$

09. පහත සඳහන් සීමා සොයන්න.

i.  $\lim_{x \rightarrow \frac{\pi}{4}} \tan x$

ii.  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

iii.  $\lim_{x \rightarrow 0} \frac{\tan 2x}{\tan 7x}$

iv.  $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

v.  $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$

vi.  $\lim_{x \rightarrow 0} \frac{1 + x - \cos x}{\sin x}$

vii.  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\operatorname{cosec} x + \tan 2x}{1 + \sin x}$

viii.  $\lim_{x \rightarrow 0} \frac{\sin 3x + \sin 5x}{4x}$

ix.  $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \tan x) \sec 2x$

10. පහත සඳහන් සීමාවන් සොයන්න.

i.  $\lim_{x \rightarrow 2} \left( \frac{x^3 - 8}{x - 2} \right)$

ii.  $\lim_{x \rightarrow 3} \left( \frac{x^7 - 3^7}{x^5 - 3^5} \right)$

iii.  $\lim_{x \rightarrow 3} \left( \frac{\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{3}}}{x - 3} \right)$

iv.  $\lim_{x \rightarrow 2} \left( \frac{x^{\frac{3}{2}} - 2^{\frac{3}{2}}}{x - 2} \right)$

v.  $\lim_{x \rightarrow 3} \left( \frac{\frac{1}{x^2} - \frac{1}{4}}{(x - 2)} \right)$

vi.  $\lim_{x \rightarrow -2} \left( \frac{x^3 + 8}{x + 2} \right)$

vii.  $\lim_{x \rightarrow 1} \left( \frac{x^2}{x - 1} - \frac{1}{x - 1} \right)$

viii.  $\lim_{x \rightarrow 3} \left( \frac{x - 3}{\sqrt{x + 6} - 3} \right)$

ix.  $\lim_{x \rightarrow 2} \left( \frac{\sqrt{x + 2} - 2}{x - 2} \right)$

x.  $\lim_{x \rightarrow 0} \left( \frac{\sqrt{4 + x^2} - 2}{x^2} \right)$

xi.  $\lim_{x \rightarrow 0} \left( \frac{\sqrt{x + 5} - \sqrt{5}}{\sqrt{x + 9} - 3} \right)$

11.  $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$  බව පෙන්වන්න.

පහත සඳහන් සීමාවන් සොයන්න.

i.  $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

ii.  $\lim_{x \rightarrow 0} \frac{x + \sin x}{2x - \sin x}$

iii.  $\lim_{x \rightarrow 0} \frac{\sin 2x \cos x}{x}$

iv.  $\lim_{x \rightarrow 0} \frac{x + \tan 3x}{x + \sin 3x}$

v.  $\lim_{x \rightarrow 0} \frac{\tan x \sec x}{2x}$

vi.  $\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x^2}$

vii.  $\lim_{x \rightarrow 0} \frac{\sin x^n}{x}$

viii.  $\lim_{x \rightarrow 0} \frac{1 - \cos 2x + x}{\sin 2x}$

ix.  $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$

x.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{\cos x - \sin x}$

xi.  $\lim_{x \rightarrow 0} \frac{\tan x^\theta}{x}$

xii.  $\lim_{x \rightarrow 0} \frac{2x - \sin x}{2x + \sin x}$

xiii.  $\lim_{x \rightarrow 0} \frac{\sin 5x \cos 3x}{x}$

xiv.  $\lim_{x \rightarrow 0} \frac{x + \sin 3x}{x - \tan 3x}$

xv.  $\lim_{x \rightarrow 0} \frac{\tan 3x \sec 2x}{4x}$

xvi.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cot x - 1}{\cos 2x}$

xvii.  $\lim_{x \rightarrow 0} \frac{1 - \cos^2 2x}{x^2}$

xviii.  $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \sin 2x}{(\cos x - \sin x) \left( x - \frac{\pi}{4} \right)}$

xix.  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

12.  $\lim_{x \rightarrow \infty} \frac{1}{x} = 0, \lim_{x \rightarrow -\infty} \frac{1}{x} = 0, \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0, \lim_{x \rightarrow -\infty} \frac{1}{x^2} = 0, \dots$  ආදී සීමාවන් සාධනයෙන් පහත සඳහන් සීමා සොයන්න.

i.  $\lim_{x \rightarrow \infty} \frac{3x^2 + x - 1}{x^4 - 2x^2 + 5x}$

ii.  $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 4}{x^2 - 6x}$

iii.  $\lim_{x \rightarrow \infty} \frac{2x^4 - 3x^2 + 6x + 3}{3x^2 + 1}$

iv.  $\lim_{x \rightarrow \infty} \frac{3x - 1}{x^2 - 2x + 5}$

v.  $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 4}{2x^2 - 6x - 5}$

vi.  $\lim_{x \rightarrow \infty} \frac{x^4 + 3x^2 - x + 3}{5x^2 + 1}$



13. පහත සඳහන් අවස්ථාවලදී සීමාවන් සොයන්න.

1.  $\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{x}+2}$
2.  $\lim_{x \rightarrow \infty} \left( 2 - \frac{1}{x} + \frac{4}{x^2} \right)$
3.  $\lim_{x \rightarrow 0} \frac{4x^3 - 2x^2 + x}{3x^2 + 2x}$
4.  $\lim_{x \rightarrow \infty} \frac{x+1}{x}$
5.  $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x - 1}{x^2 + 4}$
6.  $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$
7.  $\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 12x + 20}$
8.  $\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6}$
9.  $\lim_{x \rightarrow 1} \left[ \frac{1}{1-x} - \frac{3}{1-x^3} \right]$
10.  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$
11.  $\lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x-2} - \sqrt{2}}$
12.  $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + p^2} - p}{\sqrt{x^2 + q^2} - q}$
13.  $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x} - 1}$
14.  $\lim_{x \rightarrow a} \frac{\sqrt[n]{x} - \sqrt[n]{a}}{x - a}$
15.  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x+x^2} - 1}{x}$
16.  $\lim_{x \rightarrow \infty} \left[ \sqrt{x^2 + ax} - \sqrt{x^2 - ax} \right]$
17.  $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1}}{x + 1}$
18.  $\lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{n^2}$
19.  $\lim_{n \rightarrow \infty} \frac{1^2+2^2+3^2+\dots+n^2}{n^3}$
20.  $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^3 - 32}$
21.  $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$
22.  $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$
23.  $\lim_{x \rightarrow 1} \frac{x+x^2+x^3+\dots+x^n - n}{x-1}$
24.  $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2}$
25.  $\lim_{x \rightarrow 0} \left[ \frac{\sqrt{1+x} - \sqrt{1-x}}{x} \right]$
26.  $\lim_{x \rightarrow 64} \frac{x^{\frac{1}{2}} - 8}{x^{\frac{1}{3}} - 4}$
27.  $\lim_{x \rightarrow 0} \frac{(1+mx)^{\frac{1}{3}} - 1}{x}$
28.  $\lim_{x \rightarrow 0} \frac{x}{(1+2x)^{\frac{1}{4}} - 1}$
29.  $\lim_{x \rightarrow a} \frac{\sqrt{ax} - x}{x - a}$
30.  $\lim_{x \rightarrow 3} \frac{9 - x^2}{\sqrt{3x} - 3}$
31.  $\lim_{x \rightarrow 1} \frac{x^2 - x^{-2}}{x^5 - x^{-5}}$
1.  $\lim_{x \rightarrow 0} \frac{\tan x}{x}$
2.  $\lim_{x \rightarrow \infty} x \sin \left( \frac{1}{x} \right)$
3.  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$
4.  $\lim_{x \rightarrow 0} \frac{\tan x}{\sin x}$
5.  $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$
6.  $\lim_{x \rightarrow 0} \frac{\sin 4x + \tan 9x}{5x}$
7.  $\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cot^2 x}$
8.  $\lim_{x \rightarrow 0} \frac{2x \sin x}{\sec x - 1}$
9.  $\lim_{x \rightarrow 0} \frac{\sin 3x + \sin x - 2 \sin 2x}{x \sin^2 x}$
10.  $\lim_{x \rightarrow \pi/4} \frac{\cos^5 x - \sin^5 x}{\cos 2x}$
11.  $\lim_{x \rightarrow \pi/4} \frac{1 - \sin 2x}{\cos x - \sin x}$
12.  $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$
13.  $\lim_{x \rightarrow 0} \frac{1 - \cos ax}{1 - \cos bx}$
14.  $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos x}}$
15.  $\lim_{x \rightarrow \pi/3} \frac{1 - 2 \cos x}{\sin(x - \pi/3)}$
16.  $\lim_{x \rightarrow 0} \frac{\sin(a+x) - \sin(a-x)}{x}$
17.  $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$
18.  $\lim_{x \rightarrow 0} \frac{(a+x)^2 \sin(a+x) - a^2 \sin a}{x}$
19.  $\lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{x - \pi/4}$
20.  $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x + \cos x}$
21.  $\lim_{x \rightarrow 0} \frac{1 - \cos(\sin x)}{1 - \cos^2 x}$
22.  $\lim_{h \rightarrow 0} \frac{\sin(a+3h) - 3\sin(a+2h) + 3\sin(a+h) - \sin a}{h^3}$
23.  $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sqrt{x+2} - \sqrt{2}}$